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An experiential learning-based transit route choice model using large-scale smart-card data

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Abstract
Taking learning into account when modelling passengers' route choice behaviour improves understanding and forecasting of their preferences, which helps stakeholders better design public transport systems to meet user needs. Most empirical studies have neglected the relationship between current choices and passengers' past experiences that lead to a learning process about route attributes. This study addresses this gap by using real observed choices from smart-card data to implement a route choice model that takes into account the learning process of passengers during the inauguration of a new metro line in Santiago, Chile. An instance-based learning (IBL) model is used to represent individually perceived in-vehicle travel time in the route choice model. It accounts for recency and reinforcement of experience using the power law of forgetting. The empirical evaluation uses 8 weeks of smart-card data after the introduction of the metro line. Model parameters are evaluated, and the fit and behavioural coherence achieved by the IBL route choice model is measured against a baseline model. The baseline model neglects passenger learning from experience and assumes that all passengers use only trip descriptive information in their decision-making process. The IBL route choice model outperforms the baseline model from the fourth week after the introduction of the metro line. This empirical evidence supports the notion that after the introduction of a new metro line, passengers initially rely on descriptive travel information to estimate travel times for new alternatives. After a few weeks, they begin to incorporate their own experiences to update their perceptions.

Keywords Public transportation · Route choice · Learning model · Smart-card data

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Introduction

The modelling of public transportation route choices has often neglected the dynamic nature of real-world behaviour, largely assuming that individuals have perfect information about the attributes of alternative routes (Guo 2011; Raveau et al. 2011, 2014; Raveau and Muñoz 2014; Anderson et al. 2017; Nielsen et al. 2021; Arriagada et al. 2022). However, route choice is a typical case where travellers do not have perfect information about all dynamic route attributes. Instead, they typically use experience or travel information to form a perception about dynamic attributes. This study addresses the challenge of modelling public transport passenger behaviour more realistically by incorporating passenger learning processes that shape the perceptions of route attributes.

Capturing the route choice learning process in a realistic manner is essential to effectively account for traveller preferences so as to guide stakeholders in improving public transport systems. For example, as will be shown later, the results of our study suggest that transit agencies should reinforce travel descriptive information during the first weeks of implementation of a new travel alternative and thus facilitate users' exploration of new options and enable their exploitation of the opportunities of the new system. Further, modelling the learning process allows obtaining consistent estimators of model parameters so as to properly account for the causal relations underlying the choice process, which would not be obtainable with misspecified traditional models. Furthermore, modelling the learning process allows building models that can account for, among other things, pay-off variability and hot stove effects (e.g. Tang et al. 2017), which cannot be addressed with classical models, and therefore, enlarges the range of phenomena that can be addressed and, consequently, forecasting capabilities. Learning models may suffer the common problem of missing initial observations, precluding consistent estimation of the model parameters, but this can be addressed using the method proposed by Guevara et al. (2018).

For many years, transport studies of route choice behaviour have relied on traditional surveys of state and revealed preferences. These methods often involve cross-sectional data (Guo 2011; Raveau et al. 2011, 2014; Raveau and Muñoz 2014; Anderson et al. 2017; Nielsen et al. 2021) or longitudinal data with a small number of waves (Yáñez et al. 2009; González et al. 2017). This context creates a difficult scenario for capturing the dynamic nature of how people behave and learn in real-life contexts. Emerging technologies in transport systems improve this scheme by providing information about the actual behaviour of travellers. In particular, smart-card data allow researchers and transport operators to go back in time and observe the behaviour of all passengers over time. This provides an invaluable opportunity to understand the learning process of travellers by incorporating the relationship between passengers' past experiences and their current choices.

To the best of our knowledge, all studies on the route choice behaviour of public transport passengers using smart-card data (Schmöcker et al. 2013; Yap et al. 2020; Nassir et al. 2018; Jánošíková et al. 2014; Arriagada 2022) have ignored the relationship between choices and past experiences. In other words, they have failed to account for uncertainty in route attributes, instead often considering route attributes to be static and assuming that all passengers have the same knowledge about them. The current study fills this gap by incorporating personal perceptions of travel time, which can vary over time, into a route choice model.

Extending the literature review to private transport, in contrast to the case of public transport, many studies on the learning process can be found. Some are theoretical (Horowitz 1984; Cascetta and Cantarella 1991), and others have estimated models using experimental or simulated laboratory data (Bogers and Bierlaire 2014; Lu et al. 2014; Mahmassani and Liu 1999; Tang et al. 2017). However, very few studies have used real-world data to study learning in route choice (Vacca et al. 2019; Fusco et al. 2018). This is because real-world panel data on the learning process are virtually impossible to collect using traditional methods. The current research provides a new case study of learning in route choice by using revealed preferences collected through the use of smart cards. This type of data is not only real, granular, and massive but can also be used to study the impact of all kinds of past events in the context of route choice, thus providing a rich source for behavioural analysis.

We use the case of a new metro line in Santiago, Chile, inaugurated on 3 November 2017. Using real-world and revealed preference data, we evaluate the impact of the new metro line on passenger behaviour and incorporate a learning model into a route choice model. With this model, we seek to explain how passenger preferences and behaviour change over time following the introduction of a new metro line. We then provide guidelines for designing public transport policies based on passengers' needs when a new travel alternative is introduced in a public transport system.

The rest of this paper is organized as follows. Section "Research design and purpose" describes the case study, the transit network of Santiago, Chile. We then discuss the proposed methodology and present the model estimation results. The last section draws conclusions and discusses policy and research implications.

Research design and purpose

Research context

This study was carried out using passive transport data from the multimodal public transport (PT) network in Santiago, Chile. This system serves roughly 50% of motorized trips, and it is operated by headway scheduling; therefore, lines do not have fixed time schedules. The fare system is fully integrated, with an almost flat fare between urban buses, Metro, and one urban rail service, allowing up to three trip legs within a 2-h time window. In a typical week, 3 million passengers use the system to make 25.5 million trips. The network includes seven Metro lines, more than 300 bi-directional transit lines, and one rail service.

Very detailed demand information was obtained from the Automated Fare Collection (AFC) system in Santiago (Gschwender et al. 2016). A smart card (SC) called bip! is the only accepted payment method. Passengers must validate when boarding a bus or entering a metro station, but no validation is required when exiting. Approximately 27% of passengers evade fare payment on buses. Bus stops with particularly high demand have an off-board payment system called *zona paga* (payment zone), where passengers validate when entering the bus stop area and then board any bus without further validation.

Additionally, in the Santiago PT system, all buses are equipped with GPS devices that record a timestamp and position data every 30 s. Combining SC data and Automatic Vehicle Location (AVL) data, the boarding and alighting positions are already estimated for all validations and trips (stages) associated with an origin–destination journey using the methodology proposed by Munizaga and Palma (2012). This method first identifies the boarding locations and times of transactions made for each smart-card ID during a day. Then, the alighting location and time for each transaction are estimated by minimizing a generalized travel time function between the boarding location of the

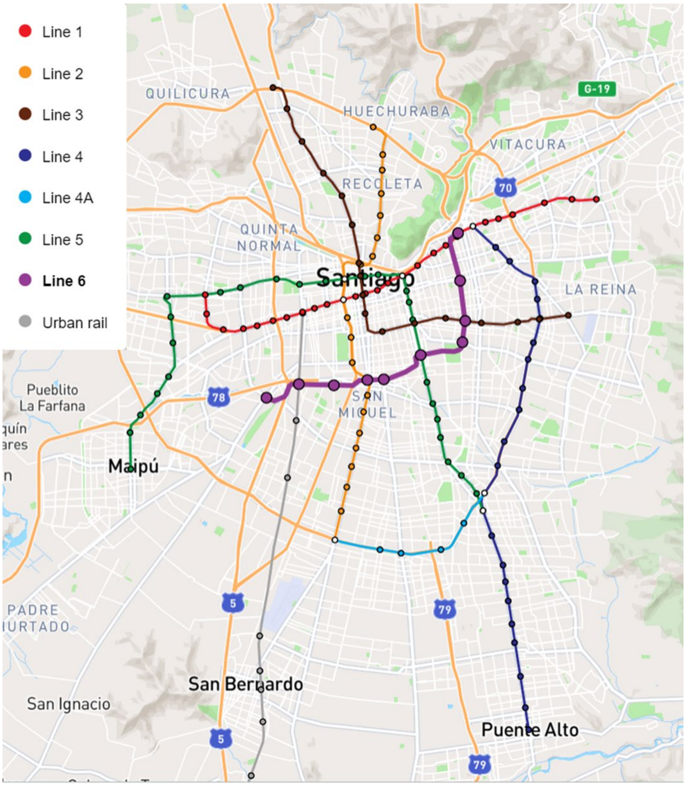

Fig. 1 Metro network and rail service in Santiago, Chile

evaluated transaction and the boarding location of the next transaction. Finally, each trip segment or transaction is associated with an origin–destination trip by using the methodology proposed by the same authors. It is important to note that Munizaga et al. (2014) reported success rates of 84.2% in estimating alighting information and 98.9% in estimating boarding locations.

In November 2017, the city of Santiago, Chile celebrated the inauguration of a new metro line (Line 6). This extension spans 15 km and introduces ten new metro stations in seven municipalities (see Fig. 1 for exact locations). Two of these municipalities thereby received their first metro station, marking a significant milestone in their transportation infrastructure. In addition, the new metro line includes three transfer stations that connect to existing metro lines, as well as a transfer station that connects to a suburban train service. The line was a completely new technology for the city at the time (autonomous trains, air conditioning, platform edge doors).

This scenario provides a unique opportunity to understand how passengers learn about new travel alternatives when they are recently introduced. We can understand how passengers change their perceptions of the travel time of previous and new alternative routes, as well as how past experiences are related to current route choices. The main purpose of this study is to model the relationship between past experiences and current choices in order to describe the learning process of passengers in the context of route choice.

Data extraction period

The study used smart-card data from a 3-month period in 2017. The data included records from October, the month prior to the launch of Line 6; November, the month when Line 6 began operating; and December, the month after Line 6 opened. From this dataset, the study focused on a subset of the SC data, specifically observations of passengers travelling during the weekday morning peak period (6:30–8:30 am). This selection aimed to retain trips likely associated with regular daily activities such as commuting to work or school.

Origin–destination pair selection

We focused on origin–destination (OD) zones where Metro Line 6 was observed as part of at least one possible trip route. The selection process involved identifying trips that used Metro Line 6 at any stage and determining their origin–destination zones within 100 ms of the corresponding origin–destination bus stops or Metro stations. We then selected those origin–destination zones where it was possible to observe at least two trips using Metro Line 6. We then selected the observed trips of users who made at least 20 trips within the same OD pair of the PT system during the study period (October, November, and December 2017).

The radius of the origin–destination zones was determined to produce the most reliable results in the models, as confirmed by sensitivity analysis. Using a larger distance for the radius would result in zones encompassing multiple stops on the same transit line. This could lead to unexpected results in the vehicle travel time coefficient. Similar results regarding the optimal radius were reported in a previous study by Arriagada et al. (2022) that used the same type of data.

Model specification

In this section, we outline the formulation of the consideration set, describe the attributes of alternative routes, and present the route choice model.

Consideration set construction

To determine the route alternatives included in the consideration set for each OD pair, we used the historical/cohort approach. This approach considers all observed routes used by passengers travelling within the specified origin and destination areas during the study period. In other words, the same consideration set was applied to all passengers for a given OD pair.

We opted for the historical/cohort approach based on the findings of Arriagada (2022), who compared the approach with heuristic approaches commonly employed in the literature on transport route choice. The results of their study indicate that the historical/cohort approach yielded models with superior predictive capabilities regarding passenger choices in the prediction sample.

A route alternative is defined by the boarding stops of the route segments, the mode (bus or metro) of the route segments, and the last alighting stop. By combining different transport lines serving the same route segment into a single mode, we were able to consider only those route alternatives that could be clearly distinguished.

It is important to note that our analysis focused on OD pairs with more than one observed route alternative after the introduction of Metro Line 6, with the goal of examining choices between at least two alternatives.

Attributes of alternative routes

Waiting time

Due to the tap-in requirement in the public transport system of Santiago, Chile, it is not possible to obtain empirical information on the arrival time of passengers at the boarding bus stop from the SC data. Consequently, it is not possible to obtain the experienced waiting time for each trip made by passengers. In this context, we estimate the waiting time for buses by using the observed frequency of buses from the AVL system, while for the metro we rely on its scheduled frequency.

To represent the waiting time, we adopted the approach of alpha divided by the frequency of vehicles, which has been employed by several authors in public transport modelling (e.g. Nguyen and Pallottino 1988; Raveau and Muñoz 2014; Schmöcker et al. 2013). In our case, we utilized the assumption of an exponential headway distribution. Therefore, the waiting time is estimated as one (alpha = 1) divided by the frequency. This choice is considered reasonable in light of previous studies that have found irregular headways in the public transport system of Santiago (Arriagada et al. 2019; Godachevich and Tirachini 2021; Arriagada et al. 2022).

Number of transfers

By applying the methodology introduced by Munizaga and Palma (2012) to link in a trip different validations within the SC data, we can derive the number of transfers for each observed trip. This number corresponds to the total number of transfers made between vehicles of the same or different travel modes.

Inertia

Inertia has a significant effect on the representation of travelers' choice behavior (Chorus and Dellaert 2009; Ben-Elia and Shiftan 2010; Vacca et al. 2019; González et al. 2017; Yáñez et al. 2009), and it can be understood as the resistance of individuals to switch from the last chosen alternative to another. In this row, a lagged dependent variable has been introduced in the utility function to capture the stickiness of each passenger to use the last chosen alternative. This variable, $ST_{ni}(t)$, takes a value of 1 if passenger n chooses the evaluated alternative i in the last trip before t and 0 otherwise.

Path size

The path size term serves as a heuristic correction to account for the correlation between routes. The original formulation was proposed by Ben-Akiva and Bierlaire (1999a), who used the fraction of correlated distance between routes. Our study has adopted a variation of this original path size correction term, proposed by Bovy et al. (2008), due to its stronger theoretical foundation. This correction term takes into account the correlated distance between route alternatives. We used this approach because the correlation arises from the geographic space where routes overlap.[1]

Equation (1) shows the expression for the path size correction (PSC) term according to Bovy et al. (2008), where l_r is the length of route section r, L_i is the total length of route i, ζ_i is the set of route sections belonging to route i, C_{OD} is the choice set of available alternative routes within the origin–destination pair OD, and δ_{rk} is the route section-route incidence number, which takes a value of 1 if route k uses route section r and a value of 0 otherwise.

$$PSC_i = \sum_{r \in \zeta_i} \frac{l_r}{L_i} \ln \frac{1}{\sum_{k \in C_{OD}} \delta_{rk}} \tag{1}$$

The path size correction term (PSC_i) belongs to the interval $(-\infty, 0]$ and takes a value of 0 when there is no overlap between the evaluated alternative and others. The value decreases as the level of route correlation increases; that is, the smaller the PSC_i value (more negative), the higher the correlation of route i. While in the case of road networks, there is a consensus that PSC reduces utility for overlapped routes, the situation is less clear for transit networks. Some studies in public transport modelling have reported contradictory findings regarding passenger perceptions of the correlation between routes. Some studies have observed a positive coefficient on the path size term, indicating a negative perception of route correlation (Hoogendoorn-Lanser and van Nes 1921; Tan et al. 2015). However, others have found a negative coefficient for the path size term, suggesting a positive perception of route correlation (Anderson et al. 2017; Arriagada et al. 2022).

The evaluation of overlap between route alternatives in this study is done at the stop level. This approach considers route alternatives that share the same route sections and modes as correlated elements. To evaluate the overlap of each route section with other alternative routes, transit line trajectories are derived from the General Transit Feed Specification (GTFS) data. However, in the case of a metro trip, where the specific trajectory chosen by passengers is unknown, it is assumed that passengers choose the route with the shortest travel time.

Average in-vehicle travel time

The in-vehicle time for each observed journey leg is determined using the AVL data. We calculate the observed in-vehicle time for each leg of a trip by calculating the difference between the estimated boarding time and the estimated alighting time. To obtain the total in-vehicle travel time for the observed trip, we sum the in-vehicle travel times of each trip leg.

[1] Some studies have used correlated travel times between route alternatives instead (Tan et al. 2015; Dixit et al. 2023). However, Dixit et al. (2023) found no significant difference in the predictive ability of the distance-based and in-vehicle travel-time-based models for the path-size correction term.

To determine the average in-vehicle travel time for a given alternative i and day t, we use all observed trips (by all passengers) made in the alternative i over the preceding days before day t. Note that this measure assigns equal weight or importance to each observed trip prior to the evaluated day.

Perceived in-vehicle travel time

This study recognizes the influence of PT passengers' perceptions, which are influenced by their previous experiences. By using SC data, we can observe passengers' repeated choices and infer their perceptions after each experience. To achieve this, we adapted the instance-based learning (IBL) model (Tang et al. 2017; Lejarraga et al. 2012) to effectively capture PT passengers' perceived in-vehicle travel time. It should be noted that this IBL model does not suffer from the initial condition problem (see Guevara et al. 2018) because full historical data is available since the opening date of the metro line.

The IBL model is based on instance-based learning theory, which was proposed to describe decision making in complex dynamic decision contexts (Lejarraga et al. 2012). This theory characterizes learning as the storage in memory of a sequence of action outcomes (instances) produced by past experiences, which are more active in memory when they are more recent and frequent (Lejarraga et al. 2012). In the context of PT route choice, an instance is a past experience of travelling along a route section using a particular mode (bus or metro) and its associated outcome, which can be a set of mode attributes.

An alternative route consists of one or more route sections (a portion of a route between two consecutive start, end, and transfer stops). Therefore, it is necessary to specify the perceived in-vehicle travel time for each route section. This is because two alternative routes may share a route section. In this case, if a passenger travels on one of the alternatives, the experienced in-vehicle travel time for the route segment common to both alternatives will influence the perceived travel time for the non-selected alternative.

Equation (2) shows, for current day t, the relative weight of the experienced in-vehicle travel time along route section r for passenger n during a past day t'. The denominator is the sum of the activation of all past experiences on route section r. In particular, the term $(t - t')^{-\delta}$ measures the recency of the experienced in-vehicle travel time and captures the rate of forgetting according to a power law. Once the relative weight of experienced in-vehicle travel time along route section r has been calculated for passenger n on all days prior to t, it is possible to calculate the perceived in-vehicle travel time of route section r on day t for passenger n by using Eq. (3). This equation is a weighted average of the perceived in-vehicle travel time of all past days that the passenger has travelled along the route section. Finally, the perceived in-vehicle travel time for an alternative route i for passenger n on day t is calculated as the sum of the perceived in-vehicle travel times along all route sections of the alternative, as shown in equation (4).

$$W_{nr}(t,t') = \frac{a_{nr}(t')(t - t')^{-\delta}}{\sum_{\tau \in H_{nr}(t)} (t - \tau)^{-\delta}} \tag{2}$$

Where:

t : current day

t': a day before day t

δ: decay parameter that captures the rate of forgetting

$H_{nr}(t)$: the set of all days before day t on which the passenger n travelled along the route section r

Table 1 Application of the IBL model in observed trips of a passenger on 6 days

Day	Experienced travel time	Perceived in-vehicle travel time			
		$\delta = 0$	$\delta = 0.5$	$\delta = 2$	$\delta = 3$
09-11-2017	12.45				
16-11-2017	10.22	12.45	12.45	12.45	12.45
20-11-2017	11.03	11.33	11.06	10.48	10.32
11-12-2017	9.63	11.23	11.18	11.05	10.99
18-12-2017	9.93	10.83	10.54	9.81	9.68
20-12-2017	11.40	10.65	10.28	9.93	9.93

Table 2 Weight of experienced travel time to calculate the perceived travel time of a specific passenger on 21-12-2017

Day	Weight of experienced travel time on day 21-12-2017			
	$\delta = 0(\%)$	$\delta = 0.5(\%)$	$\delta = 2(\%)$	$\delta = 3(\%)$
09-11-2017	17	6	0	0
16-11-2017	17	7	0	0
20-11-2017	17	7	0	0
11-12-2017	17	13	1	0
18-12-2017	17	24	10	4
20-12-2017	17	42	89	96

$a_{nr}(t')$: a binary indicator equal to 1 if the passenger n travelled trip section r on day t' and 0 otherwise.

$$T_{nr}(t) = \sum_{t' \in H_{nr}(t)} W_{nr}(t, t') X_{nr}(t') \tag{3}$$

Where:

$X_{nr}(t')$: experienced travel time along route section r for passenger n on day t'

$$PT_{ni}(t) = \sum_{r \in \zeta_i} T_{nr}(t) \tag{4}$$

Where:

ζ_i: set of route sections belonging to route i

As an illustrative example, Table 1 shows an application of the IBL model using an observed passenger in the studied database. During the period of analysis, this passenger travelled during 6 work days along a specific route section using a bus transit line. The weight of experienced in-vehicle travel time for each day is calculated following Eq. (2), and the perceived in-vehicle travel time for the route section for each day is calculated using Eq. (3). Both formulas are calculated in four cases: (i) $\delta = 0$, (ii) $\delta = 0.5$, (iii) $\delta = 2$, and (iv) $\delta = 3$. The initial perception of the bus transit line is gleaned with its first use, on November 9, i.e. 12.45 min. On November 16, the passenger had one past instance experienced, and her perception was 12.45 min for the three values of the rate of forgetting. On November 20, there were two instances, 12.45 min and 10.22 min, with a rate of forgetting equal to 0 resulting in an average value (11.33) between two previous experiences, while a rate of forgetting equal to 3 results in a value (10.32) closer to the last instance. This situation is also observed on December 11, 18, and 20, where the δ value equal to 2 or 3 obtained the perceived in-vehicle travel time closer to the last experience. These results can be explained because, as can be seen in Table 2, smaller δ values translate to higher memory activation where more distant experiences are more relevant, while greater δ values translate to lower memory activation where more recent experiences are more relevant.

Route choice model formulation

The basic model used to represent the route choice problem is the Multinomial Logit (MNL) discrete choice model. The use of the MNL model for this problem implies that correlations due to overlapping route segments are ignored because the MNL model assumes the independence of alternatives. To overcome this limitation, the analytical approach of the Path Size Logit (PSL) model is often adopted. This model accounts for correlations by introducing a deterministic term that reduces the utility of overlapping alternatives (Ben-Akiva and Bierlaire 1999b; Bovy et al. 2008). In addition, the PSL model provides a closed-form logit formula that allows straightforward estimation of fixed coefficients by maximizing the likelihood function.

In this study, we use two PSL discrete choice models to represent the passenger route choice decision process. Model 1 assumes that passengers rely solely on descriptive information about travel time in the vehicle and neglects their learning process. On the other hand, Model 2 recognizes the influence of past experience on current choices and considers that passengers have a preconceived in-vehicle travel time based on their previous use of alternative routes. The software BIOGEME (Bierlaire 2016) was used to estimate the models.

Model 1 (called the PSL model): In-vehicle travel time based on descriptive information

This model assumes that all passengers form their expectations about in-vehicle travel time based solely on descriptive information. Since the SC database does not provide information on whether travellers used descriptive travel information (or which specific traveller information system individuals used to obtain travel information), we assume that descriptive travel information from any traveller information system (such as a route planner) is generated based on historical smart-card data up to the day being evaluated. In other words, we emulate the in-vehicle travel time presented by any traveller information system as the average in-vehicle travel time for the specific alternative, derived from observed trips prior to the trip being evaluated, as explained in Sect. 3.2.

The deterministic component $V_{ni}(t)$ of Model 1 is defined in Eq. (5), where i denotes the alternative route, n represents the passenger, and t indicates the day being evaluated. The attributes for each route alternative include the average in-vehicle travel time until day t ($MeanTT_i(t)$, see Sect. 3.2), the waiting time (WT_i, see Sect. 3.2), which includes waiting time at the beginning of the trip and during transfers, the number of transfers between vehicles (TR_i, see Sect. 3.2), the stickiness to the previously chosen alternative ($ST_{ni}(t)$, see Sect. 3.2), the path size correction term (PSC_i, see Sect. 3.2), and the bus variable ($\mathrm{BUS_i}$), which takes the value of 1 if alternative i considers at least one route section by bus; otherwise, it takes the value of 0.

$$V_{ni}(t) = \beta_{TT} MeanTT_i(t) + \beta_{WT} WT_i + \beta_{TR} TR_i + \beta_{ST} ST_{ni}(t) + \beta_{PS} PSC_i + \beta_{BUS} BUS_i \quad (5)$$

Model 2 (called the IBL-PSL model): In-vehicle travel time based on learning from previous travel experiences

This model assumes that passengers form their expectations about in-vehicle travel time based on their previous travel experiences, if they have any. We postulate that passengers develop a perception of the in-vehicle travel time for each alternative route when they have utilized it in the past. To represent this perceived in-vehicle travel time, we employ the IBL model formulation, as explained in Sect. 3.2. However, if passengers have no prior experience using alternative route i, they will rely on descriptive information to inform their expectations.

In Model 2, the deterministic component $V_{ni}(t)$ is defined by Eq. (6). It includes the effect of in-vehicle travel time ($TT_{ni}(t)$) over different days and individuals. The calculation of in-vehicle travel time is based on Eq. (7), where perceived in-vehicle travel time ($PT_{ni}(t)$) is determined by Eq. (4), and the average in-vehicle travel time of the alternative ($MeanTT_i(t)$) is computed as explained in Sect. 3.2. In addition, the attribute $D_{ni}(t)$ takes the value 1 if passenger n has traveled on alternative i before time t, and 0 otherwise.

The remaining attributes in the deterministic component of Model 2 are calculated in the same way as in Model 1.

$$V_{ni}(t) = \beta_{TT} TT_{ni}(t) + \beta_{WT} WT_i + \beta_{TR} TR_i + \beta_{ST} ST_{ni}(t) + \beta_{PS} PSC_i + \beta_{BUS} BUS_i \tag{6}$$

$$TT_{ni}(t) = PT_{ni}(t) * D_{ni}(t) + MeanTT_i(t) * (1 - D_{ni}(t)) \tag{7}$$

Estimation and results

Consideration set results

After completing the consideration set construction process, we obtained a dataset of 762 OD pairs when analyzing data prior to the introduction of the Metro Line 6. Within this subset, the average number of available alternatives per OD pair was 1.76, with a maximum of 7 alternatives in the consideration sets.

After the introduction of Metro Line 6, the dataset was expanded to 1,028 OD pairs. In this dataset, the average number of available alternatives per OD pair increased to 3.06, with a maximum of 8 available alternatives within the consideration set.

Appendix A shows the distribution of the number of alternatives in OD pairs before and after the opening of the metro line. The table shows that a significant majority, more than 70%, of OD pairs have fewer than four alternatives in both cases.

Regarding the correlation between alternatives within the consideration set, we observe the patterns before and after the introduction of Metro Line 6. When analyzing the data before the introduction of the new metro line, we calculated an average path size correction term of −0.51, ranging from −1.79 to 0. On the other hand, after the introduction of Metro Line 6, the average path size correction term decreased to −0.48, with a minimum value of −1.60 and a maximum value of 0. This means that the alternatives within the consideration sets had a slightly higher correlation before the introduction of the line than afterwards.

Alternative route attributes: statistical insights

Table 3 provides statistics for each attribute of the alternatives within the consideration sets before and after the implementation of the new metro line. Similarly, Table 4 shows statistics for the alternatives chosen by passengers before and after the metro line.

In-vehicle travel time represents the average observed travel time experienced by passengers inside vehicles, including buses and the metro. Waiting time includes waiting time at the start of the trip and during transfers and is calculated as explained in Sect. 3.2. The number of transfers includes bus-to-bus, bus-to-metro, and metro-to-bus transfers. However, it is not possible to include metro-to-metro transfers in the model due to the lack of information on the specific routes used by passengers within the metro network.

The tables show that both the alternative routes within the consideration sets and the chosen alternatives showed a significant decrease in average in-vehicle travel time after the introduction of the metro line compared to before. The difference in the average in-vehicle travel time of the selected alternative routes before and after the metro line is statistically significant ($p < 0.001$), indicating a saving in in-vehicle travel time for passengers. In the period following the implementation of the metro line, a relative decrease of 10.6% in average in-vehicle travel time was observed.

Regarding waiting time, both the alternative routes within the consideration sets and the chosen alternatives experienced a slight increase after the metro line was introduced, compared to before. However, the difference in the average waiting time of the

Table 3 Statistics on the attributes of alternatives within consideration sets

Attribute	Mean	Std	Min	Max
Before metro line 6				
In-vehicle travel time (min)	47.42	16.50	12.04	116.58
Waiting time (min)	6.25	3.95	2.07	36.76
Number of transfers	0.76	0.50	0.00	2.00
After metro line 6				
In-vehicle travel time (min)	46.55	18.36	5.43	117.33
Waiting time (min)	7.82	4.25	2.07	36.76
Number of transfers	0.89	0.46	0.00	2.00

Table 4 Statistics on the attributes of the alternatives chosen by passengers

Attribute	Mean	Std	Min	Max
Before metro line 6				
In-vehicle travel time (min)	45.39	15.60	12.92	116.45
Waiting time (min)	5.36	3.51	2.07	36.76
Number of transfers	0.61	0.52	0.00	2.00
After metro line 6				
In-vehicle travel time (min)	40.58	16.56	5.43	109.73
Waiting time (min)	5.98	3.90	2.07	36.76
Number of transfers	0.57	0.53	0.00	2.00

selected alternative routes before and after the metro line is not statistically significant ($p > 0.5$).

Regarding the number of transfers, Table 3 shows a slight increase in the number of transfers for the alternative routes within the consideration sets after the introduction of the metro line. However, Table 4 shows a slight decrease in the number of transfers for the selected alternatives after the metro line compared to before. Furthermore, the difference in the average number of transfers of the chosen alternative routes before and after the metro line is not statistically significant ($p > 0.5$).

Model estimates

Models 1 and 2 were estimated using observations after the introduction of the new metro line. This approach was chosen because passengers were faced with a new scenario with additional alternative routes available. However, when calculating perceived in-vehicle travel time and average in-vehicle travel time, all trip experiences from October, November, and December 2017 were considered. It is worth noting that trip observations with consideration sets containing route alternatives that had not previously been used by a passenger were excluded from the model estimation. This is because the average in-vehicle travel time with previous observations cannot be calculated in such cases.

Specifically, data from a total of 8 weeks were used for model estimation: 4 weeks of November and 4 weeks of December. These observations correspond to trips made during 35 weekdays (excluding holidays, Saturdays, and Sundays) in November and December 2017. The trips were made by a group of 1,955 frequent passengers.

Model estimates using data from November 2017

Appendix B shows the estimation of both types of models: PSL and IBL-PSL, for each week in November.

The IBL-PSL and PSL models reported statistically significant parameters ($p < 0.001$) with the expected sign for the in-vehicle travel time, number of transfers, and stickiness parameters. The waiting time and bus parameters are statistically significant (at the usual 5% threshold) with the expected sign in both models (PSL and IBL-PSL) using data from the last 3 weeks of November. However, in the week immediately following the opening of the new metro line, waiting time is weakly significant at the 10% threshold, and the bus parameter is not statistically significant. It may be the case that during this week, people tended to increase their exploration of the new alternatives in the PT system (Erev and Barron 2005), even if they involved a higher waiting time and more trip legs on a bus. The path size correction term is weakly significant in the IBL-PSL and PSL models with data from the second week ($p < 0.10$ and $p < 0.05$, respectively). For other weeks, the path size correction term is not statistically significant. Finally, the forgetting rate is not statistically significant in all IBL-PSL models for November.

Comparing the results of the IBL-PSL and PSL models, they show a similar model fit for all evaluated weeks of November. In particular, for week 2, the PSL model shows a slightly better fit than the IBL-PSL model, while for week 4, the IBL-PSL model shows a slightly better fit than the PSL model. Consequently, we can hypothesize that during the first weeks after the introduction of a new metro line in the PT system, passengers do not rely on their previous experience. Instead, they mainly use descriptive information, represented in this study by the average in-vehicle travel time, to make route choice decisions.

Model estimates using data from December 2017

The estimates of both types of models (PSL and IBL-PSL) using observations from each week in December are reported in Appendix C.

The PSL and IBL-PSL models report statistically significant parameters with the expected sign, except for the path size parameter, which is not statistically significant for the models of all weeks in December.

Unlike the results obtained with data from November, the IBL-PSL models estimated with data from December show that the rate of forgetting parameter starts to be statistically significant ($p < 0.05$ for models in weeks 5 and 7, and $p < 0.10$ for models in weeks 6 and 8). This means that during this period, passengers started to evaluate their previous travel experiences with different levels of importance to form an in-vehicle travel time perception of the alternative. In other words, passengers started to use their previous experience to make route choice decisions.

Comparing the results of the IBL-PSL models and the PSL models during the last week of November and all weeks of December, the IBL-PSL model shows a better model fit. Consequently, during this period (weeks 4, 5, 6, 7, and 8), the inclusion of perceived in-vehicle travel time in the model specification provides greater explanatory power than models that do not account for the variation in passengers' perception of in-vehicle travel time. These results lead us to hypothesize that a few weeks after the introduction of a new metro line in a PT system, passengers begin to use their perceptions from past experiences to make route choice decisions.

Model estimates using data from November and December 2017

In summary, our results suggest that during the first weeks of a new context in a PT system, passengers' route choice behavior is different from their route choice behavior for some weeks afterwards, when they start to consider experience-based information to make route choice decisions. To prove this, we have estimated a constrained model, in which the parameters of the 8 evaluated weeks are forced to be the same (see Appendix D), and an unconstrained model, in which the parameters of the first, second, and third weeks of November can be different from the parameters of the last 5 evaluated weeks (see Table 5). The results are similar to the previous analysis regarding the sign and the statistical significance of the parameters.

Using the log-likelihood of the constrained IBL-PSL model and the log-likelihood of the unconstrained IBL-PSL model, the application of a formal likelihood ratio test largely rejects the null hypothesis that the models are equal ($p - value < 1\%$). This shows that during the first, second, and third weeks of November, passengers evaluate all their experiences at the same level of importance for a route choice decision, while for the last 5 evaluated weeks, more recent experiences became more relevant for PT passengers' route choice decisions.

The memory decay parameter (δ) is estimated to be 3.3 during the last 5 evaluated weeks. This value indicates that the most recent experience is more important than previous (distant) experiences, resembling, to some degree, the "model-free" or habit system

Table 5 Unconstrained models estimates (t tests)

Parameters	First 3 weeks		Last 5 weeks	
	IBL-PSL model	PSL model	IBL-PSL model	PSL model
In-vehicle travel time	−0.071 (−16.5)***	−0.073 (−16.7)***	−0.072 (−24.1)***	−0.072 (−22.9)***
Rate of forgetting (δ)	0.38 (0.8)		3.272 (5)***	
Waiting time	−0.06 (−8.4)***	−0.059 (−8.4)***	−0.059 (−12.5)***	−0.057 (−12.1)***
N° transfers	−1.387 (−12.5)***	−1.402 (−12.6)***	−1.463 (−20.1)***	−1.491 (−20.4)***
Inertia	2.401 (60.2)***	2.398 (60.1)***	2.553 (93.1)***	2.605 (95.7)***
PSC	0.117 (1.2)	0.14 (1.4)	−0.035 (−0.5)	−0.039 (−0.6)
Bus constant	−0.791 (−4.7)***	−0.762 (−4.5)***	−0.83 (−6.8)***	−0.789 (−6.4)***
Log-likelihood	−2816.3	−2814.0	−6213.5	−6253.2
Adjusted rho-square	0.729	0.73	0.756	0.755
AIC	5646.5	5640.0	12441.1	12518.4
BIC	5697.4	5683.6	12498.0	12567.2
N° observations	10610		25055	
N° of days	13		22	
N° of passengers	1597		1950	

+ p <0.10, *p <0.05, **p <0.01, ***p <0.001. All columns show t-values between parentheses. AIC = Akaike Information Criterion. BIC = Bayesian Information Criterion

that, according to neuroscience, controls decisions (Henriquez-Jara et al., 2023). We found a higher memory decay parameter than the values reported by Tang et al. (2017) using experimental data in a car route choice context. However, Lejarraga et al. (2012) reported a δ value of 5 in a laboratory experiment. On the other hand, some studies that have expressed perceived travel time by an exponential Markov process have also obtained a high exponential decay factor for travelers' prior experience (Vacca et al. 2019; Lu et al. 2014). Most studies that have analyzed the learning process of individuals are based on experimental data, which may be a source of difference between the results of studies that have used revealed preference data and studies that have used simulated or experimental data. We suggest that more empirical studies be conducted to understand the forgetting rates of individuals in different real-world scenarios.

Discussion

In this section, we analyse the trends in average in-vehicle travel time, perceived in-vehicle travel time, and actual travel time of observed trips.

Let's focus on a specific trip for a particular passenger. The average in-vehicle travel time is calculated as the mean of all observed travel times (from all passengers), giving equal importance to each observation, for the specific alternative up to the evaluated trip (see Sect. 3.2). In this manner, we aim to simulate the information that a passenger could receive from a transportation information tool (such as Google Maps) before making the trip. The perceived in-vehicle travel time represents the passenger's perception of the alternative, constructed using all previously observed trips made by that specific passenger in the given alternative, with more weight given to recent experiences (Eq. 4). It is important

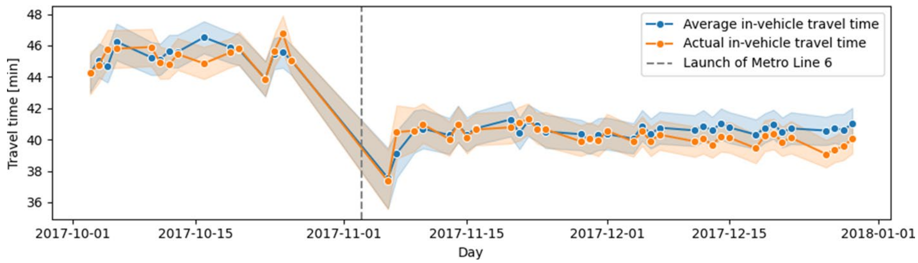

Fig. 2 The tendency of average in-vehicle travel time (used as a proxy for descriptive travel time information) and the tendency of actual in-vehicle travel time over time

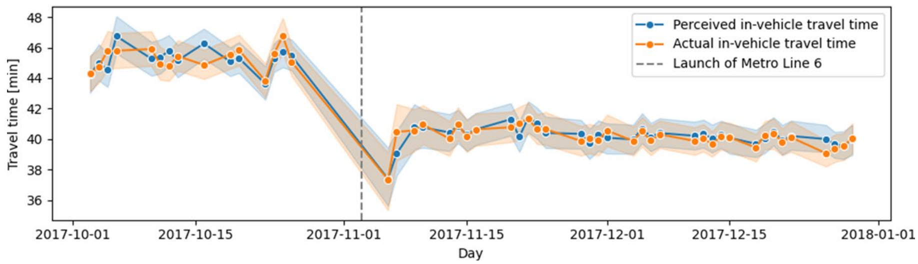

Fig. 3 The tendency of perceived in-vehicle travel time (using a rate of forgetting of 3.3) and the tendency of actual in-vehicle travel time over time

to note that both average and perceived in-vehicle travel times are the type of information that passengers can have prior to the trip. On the other hand, the actual in-vehicle travel time is the travel time that the passenger experienced during the trip. This information remains unknown to the passenger until the trip is completed.

Comparing the trends of the three measures: average in-vehicle travel time, perceived in-vehicle travel time, and actual in-vehicle travel time (see Figs. 2 and 3), there are no differences between the trends of average travel time and perceived travel time in November. However, in December, the perceived travel time is closer to the actual travel time than the average travel time. In other words, in December, the actual travel time is better represented by the perceived travel time than by the average travel time.

It is possible that during November, which was an exploration period as the new metro line was launched, the average in-vehicle travel time, which assigns the same importance to all observations, serves as an accurate predictor of the actual in-vehicle travel time. However, in December, when passengers became more adapted to the new transportation system, utilizing the same weight for all observations (including observations from November) is not as effective as using passenger perceptions, where last experience is more important than distant previous experiences.

Conclusions

In this study, the IBL model proposed by Tang et al. (2017) was applied to represent the perceived in-vehicle travel time of passengers in a large-scale multimodal PT system. The results suggest that during the first weeks of a new context in a PT system, passengers mainly use descriptive information (represented by the average in-vehicle travel time in this study) to choose a route, while a few weeks later, they start to consider experience-based information to make route choice decisions. In summary, one of the most important empirical findings of this study is that passengers use information from past experiences to select a route alternative after some initial experiences. The higher value of the memory decay parameter found using data collected a few weeks after the introduction of the new metro line means that passengers choose their routes based mostly on recent experiences rather than more distant experiences. In other words, the most recent experience is very important in passenger perceptions.

These results provide potentially valuable guidelines for policy makers, authorities, and operators in the design and regulation of public transport systems. The study suggests that it is highly recommended that transit authorities reinforce trip descriptive information during the first weeks of implementation of a new travel alternative. This information allows passengers to form their initial perceptions. However, when providing travellers with a travel time estimate, a balance between descriptive (pre-trip information) and experiential (actual travel time) information is necessary, as passengers begin to use their own experience as they become more familiar with the new transportation context.

This research presents a new case study using smart-card data consisting of real-world data from the public transport system of Santiago, Chile, to apply the IBL model, a traveller learning model. This highlights the valuable opportunities offered by passive transportation data to study the behavior of large numbers of passengers over time. This approach significantly reduces both the time and cost associated with collecting information on passenger preferences and route choices in public transport. By incorporating actual user evaluations of system attributes, this methodology facilitates more informed transportation decision making and improves the design and operation of public transportation services.

Several challenges remain to be addressed in modelling the learning process of passengers. First, the current study only considers perceived travel time in the vehicle as an attribute that evolves over time. We recognize that waiting time is also a dynamic attribute that can vary from day to day and between individual travel experiences. However, accurate estimation of the IBL model for the waiting time attribute requires access to data on the actual waiting time experienced by passengers during each trip. Unfortunately, the data available in this study, such as the SC data, do not provide this information as they do for in-vehicle travel time. Future research should aim to address this limitation by using appropriate data sources to capture the variability of waiting times experienced by passengers.

Second, this study uses data processed with the methodology proposed by Munizaga and Palma (2012) for estimating boarding, alighting, and transit line information on each validation. Although this methodology has demonstrated considerable accuracy in previous studies (Munizaga and Palma 2012; Munizaga et al. 2014), further research is necessary to investigate the impact of potential estimation errors in boarding, alighting, and transit line information on route choice models that use smart-card data.

Another potential avenue of research is to examine the choice set by using an individual approach to identifying its alternatives. In our study, the choice set for each OD pair was determined based on the choices made by all passengers travelling between those pairs. Consequently, the choice set remained the same for all passengers within each OD pair. Future research could examine individual consideration sets constructed by observing the choices made by each passenger over a period of time. This approach would allow us to determine whether the use of individual consideration sets yields results differing from those of our study.

Distribution of number of alternatives in origin–destination pairs

See Table 6.

Table 6 Distribution of number of alternatives in OD pairs

N° of alternatives	N° of OD pairs before L6	N° of OD pairs after L6
1	416	0
2	194	471
3	95	259
4	41	145
5	11	95
6	3	38
7	2	17
8	0	3

Estimation of PSL and PSL-IBL using 4 weeks of data from November 2017

See Table 7.

Table 7 PSL model and PSL-IBL model estimates (t tests) using 4 weeks of data from November 2017

Parameters	IBL-PSL model	PSL model
Week 1: November 06, 07, 09, and 10		
In-vehicle travel time	−0.082 (−9.1)***	−0.083 (-9)***
Rate of forgetting (δ)	0.306 (0.2)	
Waiting time	−0.028 (−1.9)$^{+}$	−0.027 (−1.8)$^{+}$
N° transfers	−1.917 (−7.6)***	−1.931 (−7.7)***
Inertia	2.144 (23.8)***	2.14 (24.5)***
PSC	0.179 (0.8)	0.191 (0.8)
Bus constant	−0.548 (−1.6)	−0.523 (−1.5)
Log-likelihood	−578.7	−579.7
Adjusted rho-square	0.704	0.704
AIC	1171.4	1171.5
BIC	1211.0	1205.4
N° observations	2127	
N° of days	4	
N° of passengers	964	
Week 2: November 13, 14, 15, and 16		
In-vehicle travel time	−0.063 (−8.4)***	−0.067 (−8.7)***
Rate of forgetting (δ)	0.244 (0.4)	
Waiting time	−0.078 (−6.2)***	−0.078 (−6.2)***
N° transfers	−1.134 (−5.8)***	−1.164 (−5.9)***
Inertia	2.392 (34.2)***	2.39 (34.2)***
PSC	0.311 (1.8)+	0.337 (2)*
Bus constant	−1.013 (−3.3)**	−0.962 (−3.1)**
Log-likelihood	−930.1	−927.1
Adjusted rho-square	0.724	0.725
AIC	1874.1	1866.2
BIC	1917.1	1903.1
N° observations	3449	
N° of days	4	
N° of passengers	1306	
Week 3: November 20, 21, 22, 23, and 24		
In-vehicle travel time	−0.069 (−10.7)***	−0.072 (−10.8)***
Rate of forgetting (δ)	0.51 (0.6)	
Waiting time	−0.06 (−5.7)***	−0.059 (−5.6)***
N° transfers	−1.333 (−8.3)***	−1.34 (−8.3)***
Inertia	2.532 (42.6)***	2.528 (42.6)***
PSC	−0.032 (−0.2)	−0.011 (−0.1)
Bus constant	−0.716 (−2.8)**	−0.701 (−2.8)**
Log-likelihood	−1290.9	−1291.0

Table 7 (continued)

Parameters	IBL-PSL model	PSL model
Adjusted rho-square	0.743	0.743
AIC	2595.9	2594.0
BIC	2641.5	2633.2
N° observations	5034	
N° of days	5	
N° of passengers	1568	
Week 4: November 28, 29, and 30, and December 1		
In-vehicle travel time	−0.066 (−9.2)***	−0.068 (−9.2)***
Rate of forgetting (δ)	1.326 (1.6)	
Waiting time	−0.047 (−4.1)***	−0.047 (−4.1)***
N° transfers	−1.398 (−7.6)***	−1.401 (−7.6)***
Inertia	2.586 (39.3)***	2.591 (39.4)***
PSC	0.019 (0.1)	0.055 (0.3)
Bus constant	−0.879 (−3.1)**	−0.855 (-3)**
Log-likelihood	−1062.5	−1065.3
Adjusted rho-square	0.756	0.755
AIC	2139.0	2142.6
BIC	2183.6	2180.8
N° observations	4332	
N° of days	4	
N° of passengers	1616	

+ p <0.10, * p <0.05, ** p <0.01, *** p <0.001

All columns show t-values between parentheses. AIC = Akaike Information Criterion and BIC = Bayesian Information Criterion

Estimation of PSL and PSL-IBL models using 4 weeks of data from December 2017

See Table 8.

Table 8 PSL model and PSL-IBL model estimates (t tests) using 4 weeks of data from December 2017

Parameters	IBL-PSL model	PSL model
Week 5: December 4, 5, 6, and 7		
In-vehicle travel time	−0.074 (−10.6)***	−0.075 (−10.4)***
Rate of forgetting (δ)	1.718 (2.5)*	
Waiting time	−0.059 (−5.3)***	−0.058 (−5.3)***
N° transfers	−1.35 (−7.9)***	−1.361 (−7.9)***
Inertia	2.523 (40.1)***	2.54 (40.5)***
PSC	0.003 (0)	0.021 (0.1)
Bus constant	−0.891 (−3.2)**	−0.868 (−3.1)**
Log-likelihood	−1164.9	−1168.6
Adjusted rho-square	0.747	0.746
AIC	2343.8	2349.1
BIC	2388.8	2387.7
N° observations	4580	
N° of days	4	
N° of passengers	1683	
Week 6: December 11, 12, 13, 14, and 15		
In-vehicle travel time	−0.076 (−11.8)***	−0.074 (-11)***
Rate of forgetting (δ)	4.434 (1.8)$^{+}$	
Waiting time	−0.054 (−5.3)***	−0.05 (-5)***
N° transfers	−1.583 (−10.2)***	−1.605 (−10.4)***
Inertia	2.613 (44.8)***	2.679 (46.3)***
PSC	−0.081 (−0.6)	−0.09 (−0.6)
Bus constant	−0.776 (−2.9)**	−0.748 (−2.8)**
Log-likelihood	−1375.4	−1386.6
Adjusted rho-square	0.765	0.763
AIC	2764.8	2785.2
BIC	2811.4	2825.2
N° observations	5782	
N° of days	5	
N° of passengers	1796	
Week 7: December 18, 19, 20, 21, and 22		
In-vehicle travel time	−0.077 (-12)***	−0.074 (−11.1)***
Rate of forgetting (δ)	2.906 (2.7)**	
Waiting time	−0.071 (−7.2)***	−0.068 (−6.9)***
N° transfers	−1.564 (−10.2)***	−1.577 (−10.3)***
Inertia	2.54 (44.1)***	2.608 (45.7)***
PSC	0.068 (0.5)	0.05 (0.4)
Bus constant	−0.661 (−2.5)*	−0.636 (−2.4)*

Table 8 (continued)

Parameters	IBL-PSL model	PSL model
Log-likelihood	−1415.1	−1428.4
Adjusted rho-square	0.759	0.756
AIC	2844.2	2868.8
BIC	2890.8	2908.7
N° observations	5738	
N° of days	5	
N° of passengers	1809	
Week 8: December 26, 27, and 29		
In-vehicle travel time	−0.072 (−10.2)***	−0.066 (−9.3)***
Rate of forgetting (δ)	6.683 (1.8)$^{+}$	
Waiting time	−0.066 (−6.1)***	−0.063 (−5.9)***
N° transfers	−1.424 (−9.1)***	−1.472 (−9.3)***
Inertia	2.484 (39)***	2.585 (41.3)***
PSC	−0.193 (−1.3)	−0.239 (−1.6)
Bus constant	−0.959 (−3.4)***	−0.893 (−3.1)**
Log-likelihood	−1186.4	−1198.5
Adjusted rho-square	0.749	0.747
AIC	2386.8	2408.9
BIC	2431.9	2447.5
N° observations	4623	
N° of days	4	
N° of passengers	1770	

+ p <0.10, * p <0.05, ** p <0.01, *** p <0.001

All columns show t-values between parentheses. AIC = Akaike Information Criterion and BIC = Bayesian Information Criterion

Estimation of PSL and PSL-IBL models using all weeks of data

See Table 9.

Table 9 Constrained models estimates (t tests)

Parameters	All weeks	
	IBL-PSL model	PSL model
In-vehicle travel time	−0.071 (−29.3)***	−0.072 (−28.5)***
Rate of forgetting (δ)	2.587 (5.7)***	
Waiting time	−0.06 (−15.2)***	−0.058 (−14.9)***
N° transfers	−1.438 (−23.7)***	−1.471 (−24.2)***
Inertia	2.51 (111.4)***	2.543 (113.3)***
PSC	0.007 (0.1)	0.013 (0.2)
Bus constant	−0.821 (−8.3)***	−0.773 (−7.8)***
Log-likelihood	−9041.4	−9077.8
Adjusted rho-square	0.748	0.747
AIC	18096.7	18167.6
BIC	18156.1	18218.5
N° observations	35665	
N° of days	35	
N° of passengers	1955	

+p<0.10, * p <0.05, ** p<0.01, ***p <0.001

All columns show t-values between parentheses. AIC = Akaike Information Criterion. BIC = Bayesian Information Criterion

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